

VIRRAT AIJANNEVA, Finland

WMO: 027350

Lat: 62.3278N

Lon: 23.5422E

Elev: 458

StdP: 14.45

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-9.7	-20.3	1.8	-15.7	-13.9	2.6	-9.6	20.8	27.0	18.9	26.8	0.9	160	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	80.1	65.1	76.4	63.2	73.1	61.7	67.6	76.0	65.4	73.1	63.4	70.5	7.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	93.2	70.9	62.4	85.7	68.7	60.3	79.4	66.4	32.3	76.2	30.6	72.7	29.0	70.7	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	16.2	14.4	DB	-20.6	83.0	5.4	5.3	-24.4	86.9	-27.6	90.0	-30.6	92.9	-34.5	96.8	(4)
(5)			WB	-20.8	70.1	5.3	3.6	-24.7	72.7	-27.8	74.8	-30.8	76.9	-34.6	79.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.9	17.4	20.7	26.0	36.9	48.3	55.0	60.4	57.3	49.6	38.1	31.9	24.2		
	DBStd	16.80	12.44	12.46	9.75	6.54	7.95	6.24	6.01	5.31	6.08	6.85	8.48	11.28		
	HDD50	4916	1009	821	743	393	126	20	2	6	79	372	544	801		
	HDD65	9562	1474	1241	1208	843	519	305	170	246	462	834	994	1266		
	CDD50	871	0	0	0	1	73	171	324	232	67	3	0	0		
	CDD65	42	0	0	0	0	2	6	27	7	0	0	0	0		
	CDH74	585	0	0	0	0	53	103	344	85	0	0	0	0		
	CDH80	91	0	0	0	0	2	11	62	16	0	0	0	0		
	WSAvg	6.8	6.9	7.4	7.7	6.8	6.9	6.9	5.6	6.1	6.2	6.8	6.8	7.4		
	PrecAvg	21.9	1.5	1.1	1.1	1.1	1.6	2.4	3.0	2.6	2.1	2.1	1.7	1.6		
	PrecMax	29.3	3.2	2.6	1.9	2.3	3.2	4.2	5.4	5.0	4.6	4.4	3.8	2.9		
(15) Precipitation	PrecMin	16.6	0.4	0.0	0.1	0.1	0.2	0.5	0.5	0.6	0.5	0.4	0.1	0.6		
	PrecStd	3.0	0.7	0.6	0.4	0.7	0.9	1.0	1.3	1.1	1.3	1.1	0.8	0.6		
	DB	37.9	41.7	48.8	65.6	79.3	81.5	85.2	82.6	69.5	56.9	50.7	44.3			
	MCWB	35.9	37.3	39.7	49.4	59.5	64.4	68.3	68.3	60.3	52.7	48.6	43.1			
	2% DB	35.6	38.7	43.4	58.5	74.8	77.1	81.5	75.6	65.4	53.1	46.7	39.9			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	34.0	36.0	37.8	46.9	58.0	62.9	67.0	63.9	58.1	50.7	45.4	38.5			
	5% DB	33.7	36.4	40.5	52.8	71.0	72.4	77.9	72.2	62.4	51.1	44.4	37.4			
	MCWB	33.0	35.0	36.1	43.7	56.3	60.4	65.0	61.5	56.3	49.0	43.4	36.2			
	10% DB	32.4	34.1	37.7	48.8	65.7	68.7	74.3	68.7	59.9	48.6	41.6	35.7			
	MCWB	31.6	33.2	34.3	41.7	53.7	58.5	63.9	59.9	55.5	46.8	40.9	34.9			
	0.4% WB	36.6	37.8	40.7	50.2	62.2	67.9	71.6	69.2	61.7	53.6	49.0	43.4			
	MCDB	37.5	39.9	45.0	62.3	73.8	76.3	80.7	79.1	67.0	55.7	50.4	44.1			
	2% WB	34.3	36.3	38.5	47.7	60.0	64.5	68.6	65.3	59.6	51.6	45.7	38.8			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	35.0	38.1	42.3	56.3	70.3	73.2	77.4	72.6	63.1	52.8	46.7	39.9			
	5% WB	33.1	35.1	36.7	44.8	57.6	62.2	66.9	63.2	57.9	49.4	43.4	36.4			
	MCDB	33.6	36.4	39.6	51.5	68.1	70.1	74.8	69.6	61.1	50.7	44.2	37.0			
	10% WB	31.9	33.4	34.9	42.5	55.0	59.9	65.0	61.4	56.2	46.7	41.2	34.8			
	MCDB	32.5	34.2	37.2	47.7	65.4	67.1	72.4	66.6	59.1	48.3	41.7	35.3			
	MDBR	10.8	11.7	16.2	17.0	20.9	19.3	19.7	19.8	15.9	10.8	7.0	8.6			
	5% DB	6.8	10.0	15.1	26.1	31.5	27.0	25.6	26.2	20.7	13.6	8.7	8.3			
	MCWBR	6.5	8.2	10.5	15.9	16.0	14.8	12.9	14.8	14.3	10.9	8.2	7.8			
(36) Mean Daily Temperature Range	5% WB	6.7	8.9	13.3	22.3	27.3	22.7	22.7	21.2	17.2	12.1	8.2	8.3			
	MCWBR	6.6	7.5	9.9	14.4	14.8	13.2	12.6	12.7	12.5	10.7	7.8	8.1			
	taub	0.237	0.288	0.314	0.353	0.349	0.348	0.371	0.356	0.332	0.315	0.258	0.353			
	taud	2.061	2.164	2.240	2.355	2.444	2.469	2.423	2.456	2.518	2.408	2.157	2.615			
(40) Clear-Sky Solar Irradiance	Ebn at Noon	136	208	250	264	276	278	267	262	248	203	127	85			
	Edh at Noon	11	21	28	31	31	31	32	28	22	17	10	7			
	RadAvg	62	241	719	1167	1575	1693	1596	1226	727	294	76	31			
(44) All-Sky Solar Radiation	RadStd	10	55	78	143	167	144	158	131	86	61	14	3			

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	-236	N/A	N/A	N/A

Nomenclature: See separate page